

PER ASPERA AD ASTRA (CLOUD)

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Hands-on course #8

Kandula final project

04/08/2022

OPS SCHOOL

WHO AM I

3 factors of Authentication



- Personal
  - 49-year-old
  - 32 years Israel citizen (Alia 1990)
  - Happy and free Divorces
  - 4 years (cancer runner)
  - Mom to 19-year great daughter
- Professional
  - One women show more that 10 years with full responsibility on all R&D processes and infrastructure
  - DevOps Architect & Sr. Software Engineer at CMT
  - Mentor and development evangelist
  - C++ speaker ☺







SOMETHING I AM

SOMETHING I HAVE

- Personal
  - Cat
  - Daughter
  - Sister
  - Love
- Professional
  - 28-year development experience from bottom to top (Development, Architecture, Design…)
  - Some theoretical knowledge about Cloud technology
  - Near 0 experienced knowledge of Cloud technology



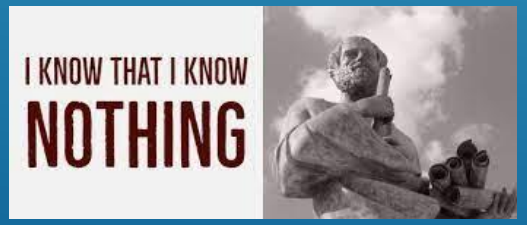
SOMETHING I KNOW

- Personal
  - **I know that I know NOTHING**
  - Hobbies : Abseiling, Hiking, sewing
  - Each day that we not smile and not cause to someone smile is wasted day
- Professional
  - Few ☺ software languages
    - C, C++, C#, Java, Perl, TCl, PHP, Python
  - There are many technology to learn









WHAT WAS MY TARGET OF COURSE

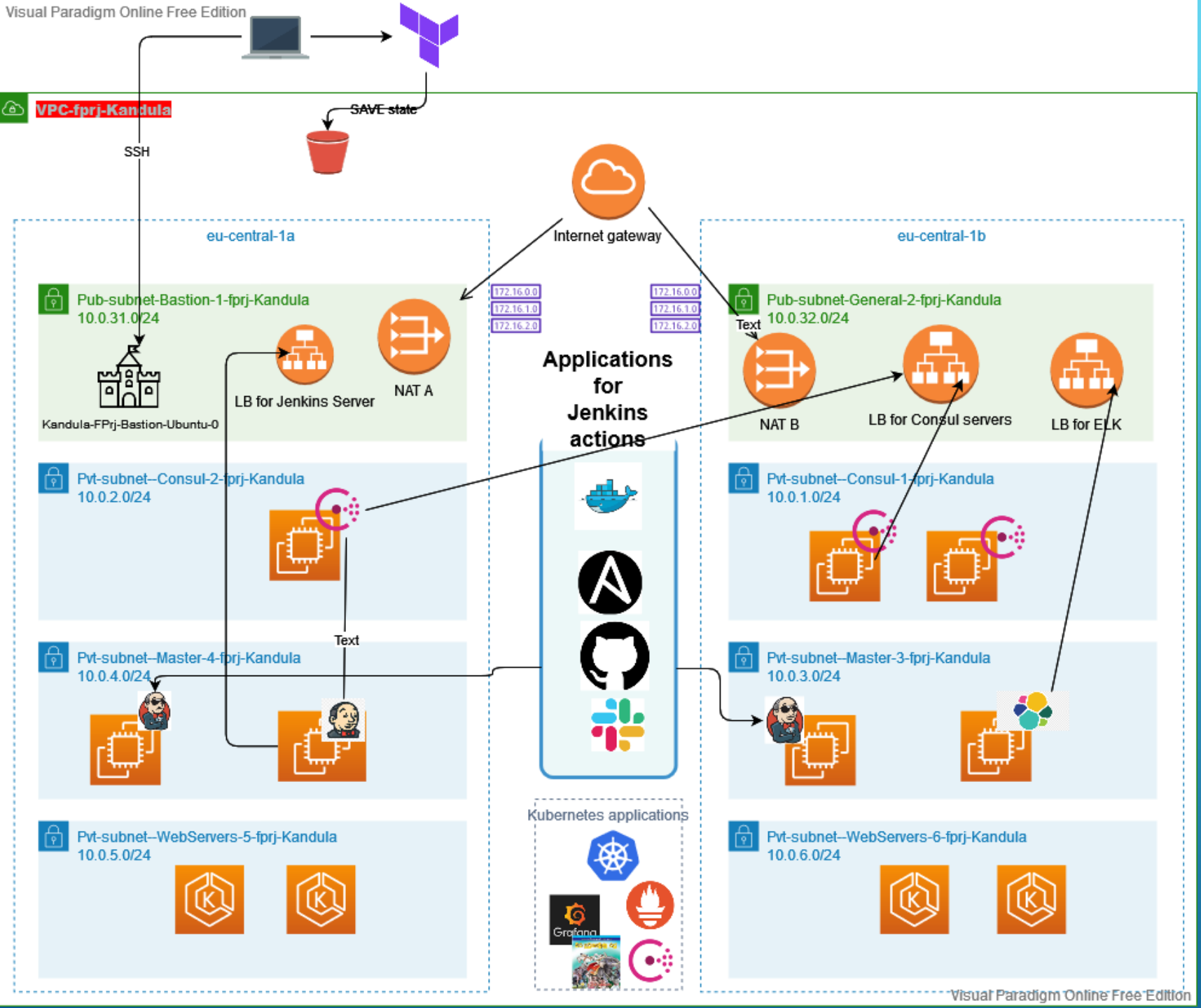
- Personal
  - Develop my self again (I do it each 5-7 years)
  - Meet new interesting people from the Hitech industry
  - Meet personal challenge and compete according my self “yesterday” only
- Professional
  - **Learn and “fill” cloud infrastructure near as possible to real usage**



LET'S START



AWS infrastructure diagram

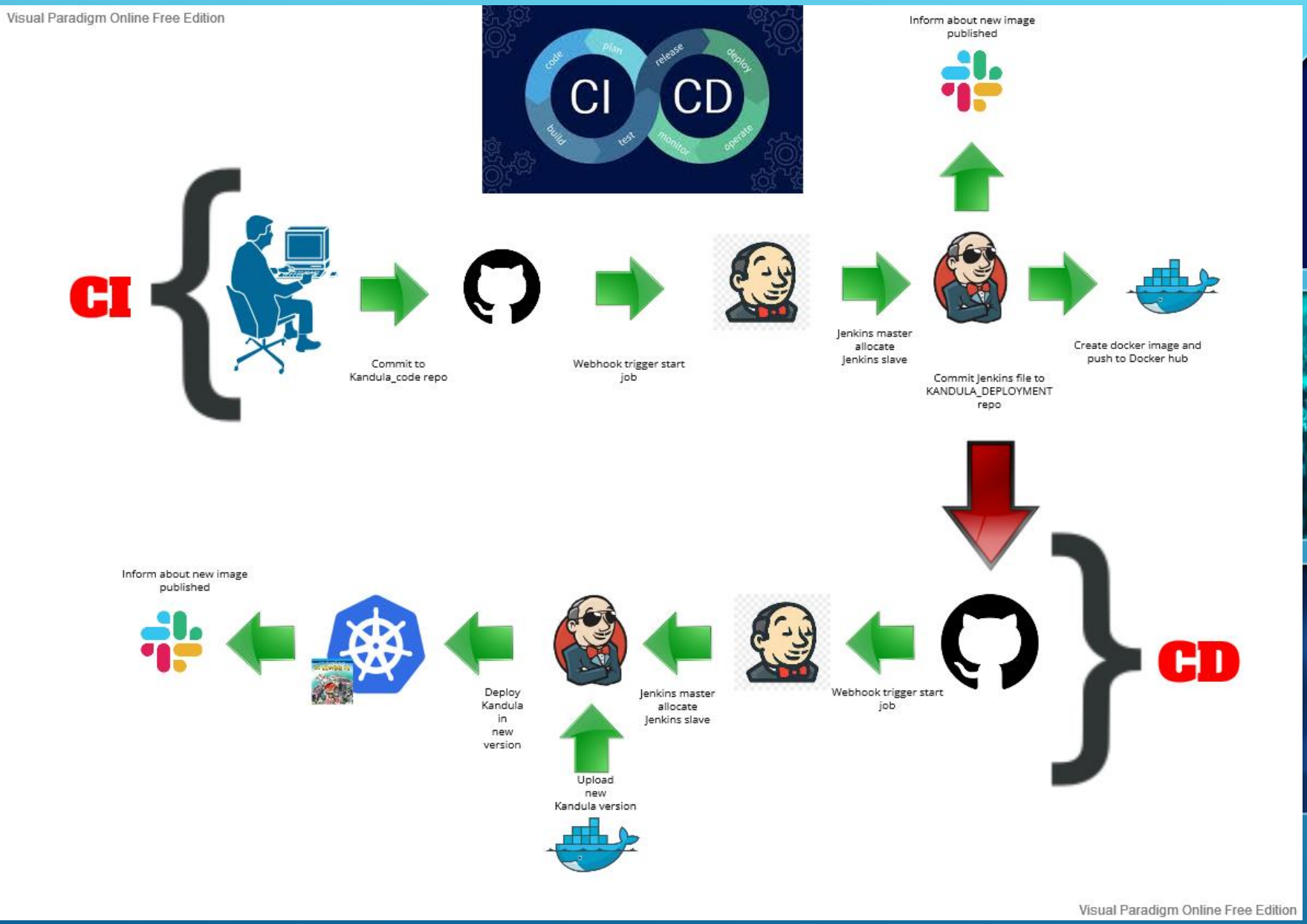








CI-CD flow


Visual Paradigm Online Free Edition
CI
CD
code
plan
release
deploy
build
test
monitor
operate
Inform about new image published
Commit to Kandula_code repo
Webhook trigger start job
Jenkins master allocate Jenkins slave
Commit Jenkins file to KANDULA_DEPLOYMENT repo
Create docker image and push to Docker hub
Webhook trigger start job
Jenkins master allocate Jenkins slave
Deploy Kandula in new version
Upload new Kandula version
Inform about new image published
Visual Paradigm Online Free Edition







DIFFERENT TOOLS AND DECISIONS

- Terraform
  - Creates all VPC/EC2/KUBERNETEST/… infrastructure for first create an environment
  - Use modules for create more modulated environment
- Ansible
  - Used from Jenkins job any time at Terraform build for first time and any time when user want update environment from Jenkins job Used for install manually
    - Consul servers and agents on EC2
    - Node Exporter
    - Filebeat
- Git
  - As Source control system







DIFFERENT TOOLS AND DECISIONS

- Shell
  - Used for create an environment
  - Used inside user data
  - Used in TF remote-exec provisioning via bastion
- Jenkins server was created from saved ami each time
- Jenkins slaves/Consul server/ELK server
  - Installed from TF user data and remote-exec provisional and user data
- HELM used for install on Kubernetes
  - Prometheus
  - Grafana
  - Consul







DIFFERENT TOOLS AND DECISIONS

- Python
  - Used for Kandula itself code
  - Used for CLI tool for Kandula
  - Used for internal scripts
- Secrets
  - Jenkins server secret system used for keep all git, docker hub, private keys secrets
  - Kubernetes secrets used for keep AWS secrets for start kandula application
- Slack
  - Used for notification system of Jenkins jobs
- Separated TF environment
  - For save time and easy create part of environment was decided to use relevant TF code for Logical parts of project
  - There exist 1 shell script which execute full create AWS environment



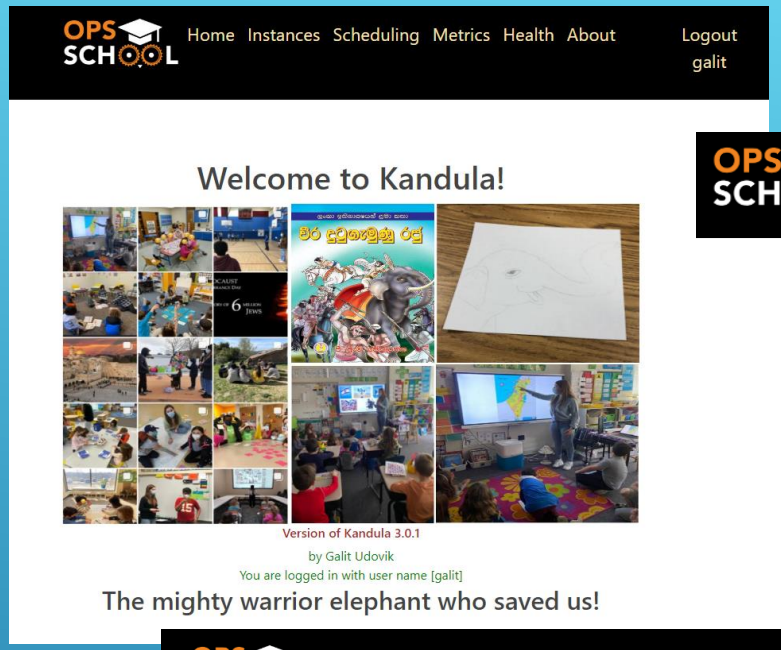
DIFFERENT TOOLS AND DECISIONS

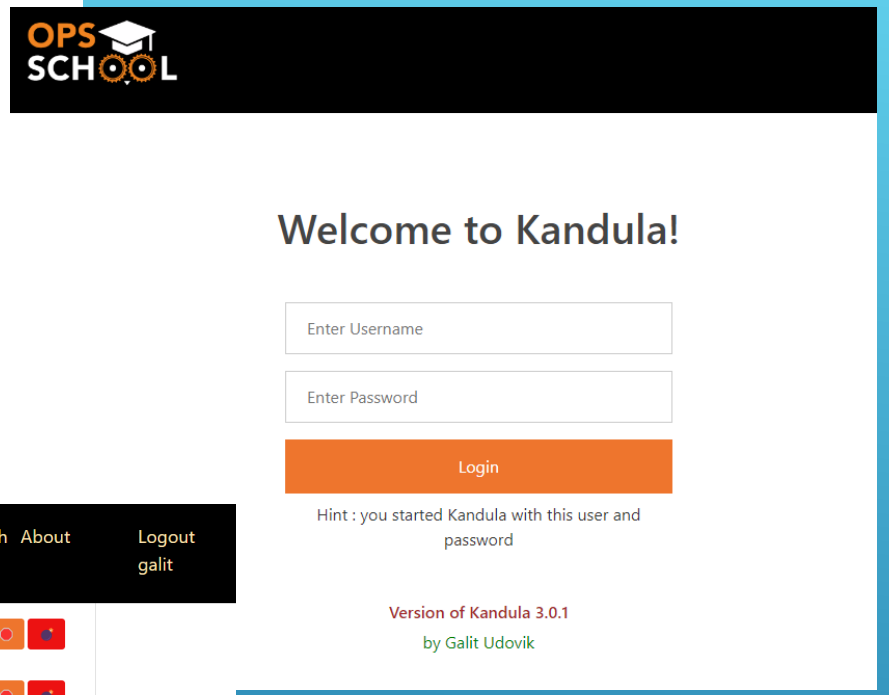
- Jenkins slaves are “self registered machines” via Jenkins CLI
- 3 Ansible Jenkins job executed at end of create TF environment
- All deployment to Kubernetes are executed via TF at beginning or yaml file via Jenkins job any time
- Kubernetes Ingress used for
  - Grafana inside KUBE
  - Kandula inside KUBE
  - Prometheus inside KUBE
- Load balancers for private EC2 machines
  - Decided to use 1 per server and not 1 for all with different targets and ports
- I purchased Domain for use its name for all AWS environment UI
- TLS termination
  - Used a public AWS certificate for all web UI of all applications

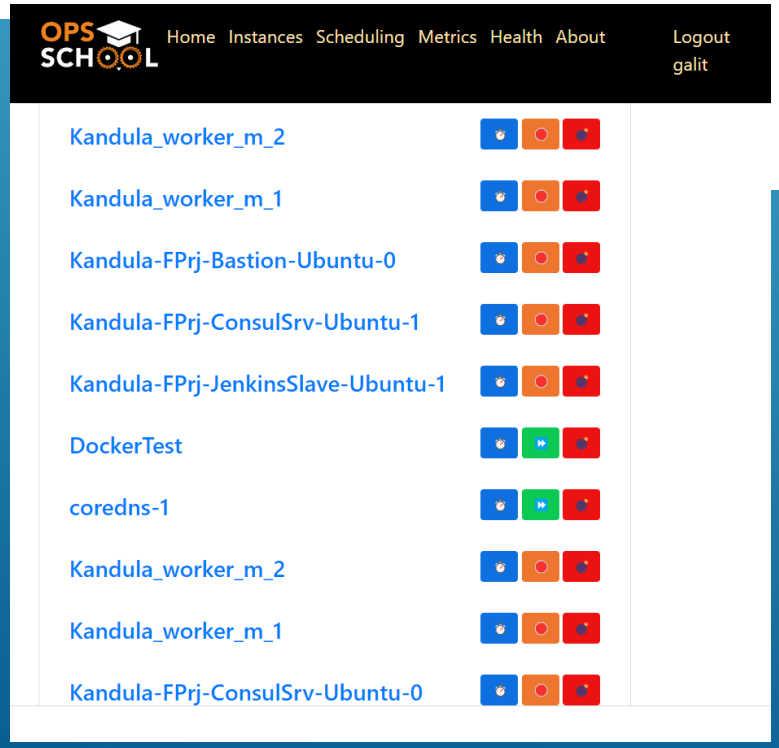


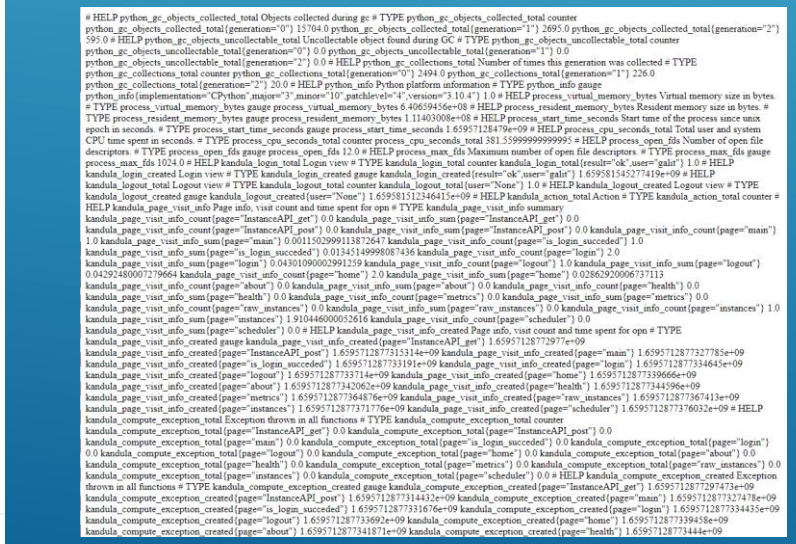
Kandula application

https://kandula.galitu.site







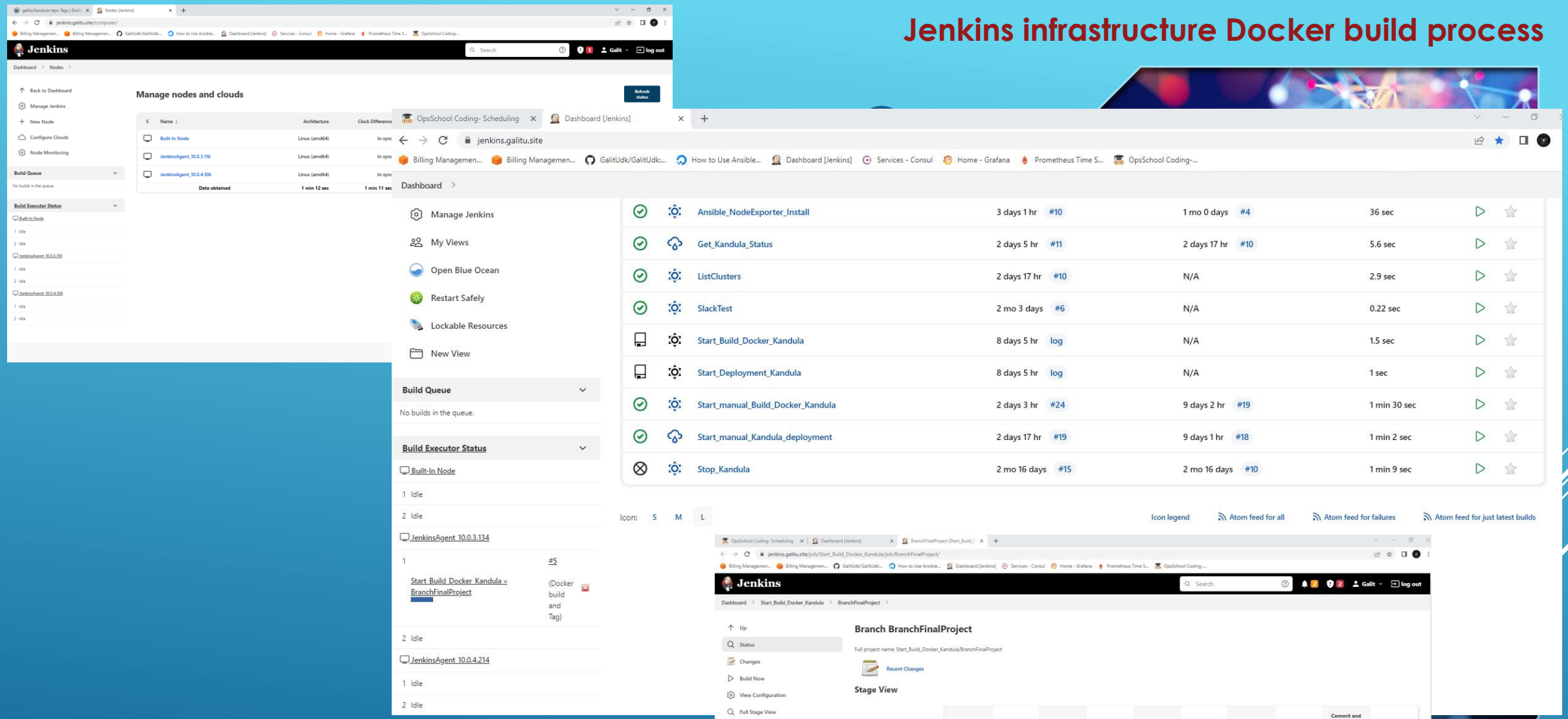




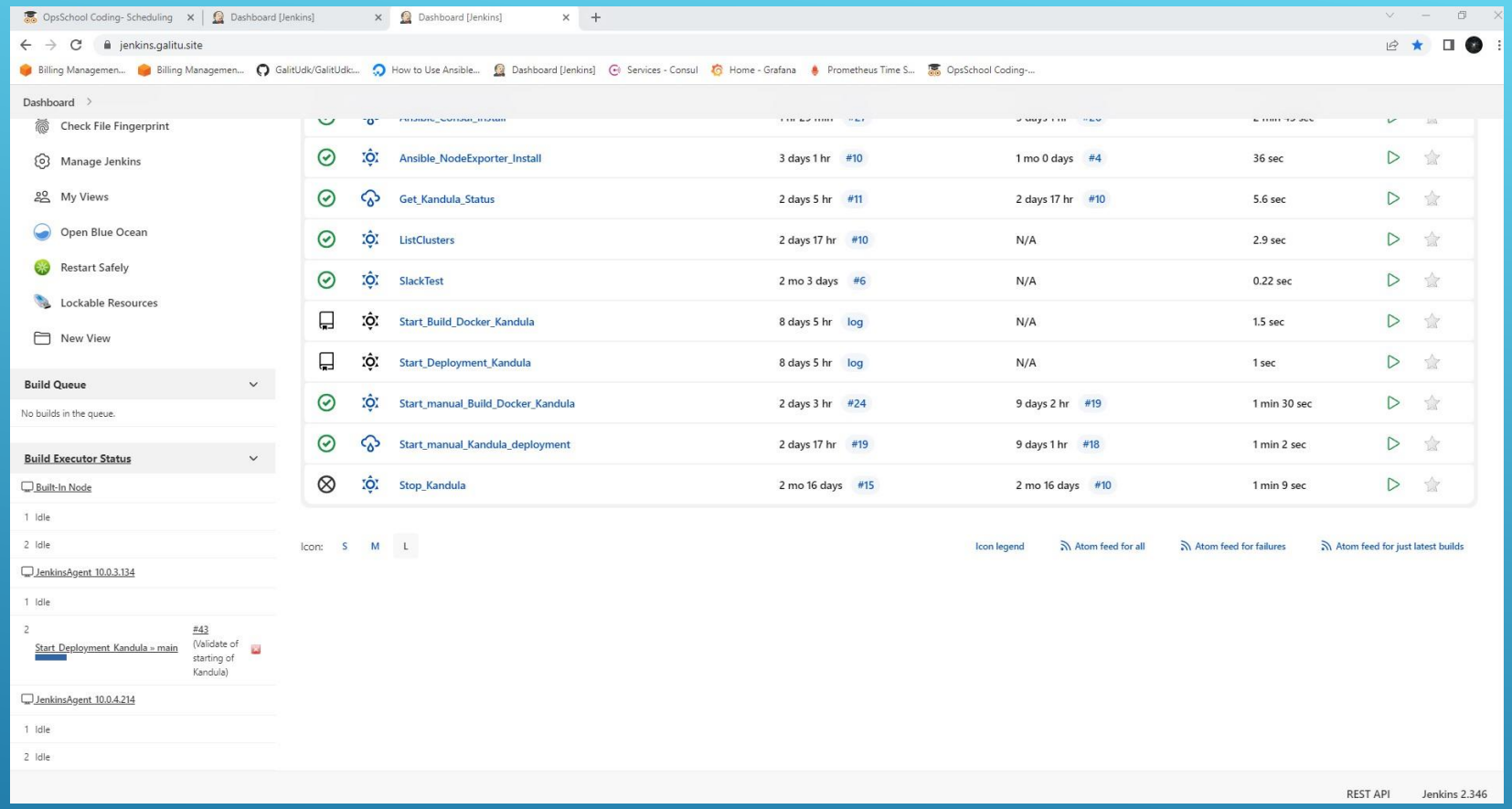




Jenkins infrastructure Docker build process



Jenkins infrastructure Deployment









Consul servers

galit_dc_kandula | Services | Nodes | Key/Value | ACL | Intentions

Services 14 total

Search

- Bastion
  1 Instance Bastion
- consul
  3 Instances
- consul-consul-dns-consul
  Registered via Kubernetes 4 Instances k8s
- ELK
  1 Instance ELK
- grafana-grafana
  Registered via Kubernetes 1 Instance k8s
- Jenkins-Server
  1 Instance jenkins
- Jenkins-Slave
  2 Instances jenkins
- kandula-application
  Registered via Kubernetes 2 Instances fprj, k8s, kandula
- kubernetes-default
  Registered via Kubernetes 2 Instances k8s
- prometheus-alertmanager-monitoring
  Registered via Kubernetes 1 Instance k8s
- prometheus-kube-state-metrics-monitoring
  Registered via Kubernetes 1 Instance k8s

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Grafana dashboards … Kubernetes









Grafana dashboards … Kubernetes









Grafana dashboards … Jenkins

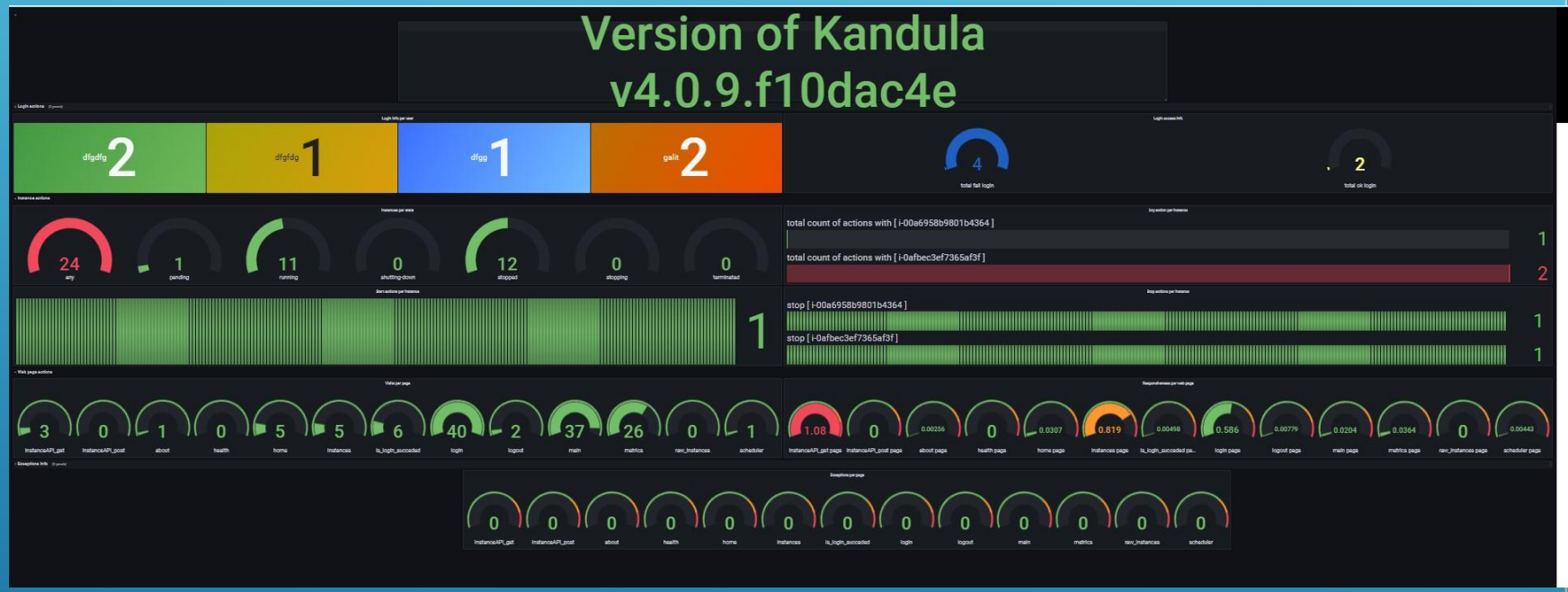








Grafana dashboards …Kandula

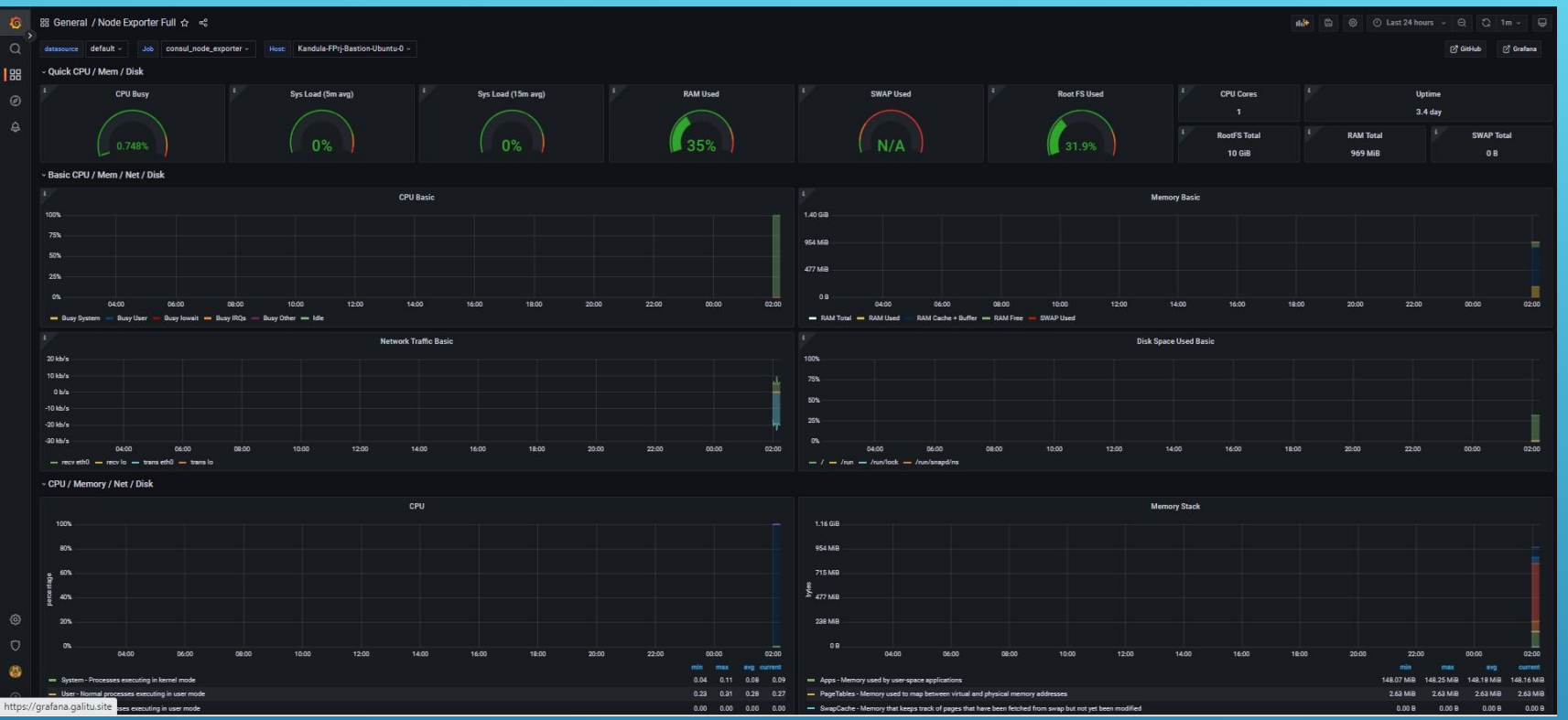








Grafana dashboards ...Node Exporter









Kandula CLI

```
$$  python kandulacli.py --help
Usage: kandulacli.py [OPTIONS] COMMAND [ARGS]...

Options:
  --help  Show this message and exit.

Commands:
  get_instance
  get_instances
  start_instance
  stop_instance
  terminate_instance
```

```
$$  python kandulacli.py get_instance --help
Usage: kandulacli.py get_instance [OPTIONS]

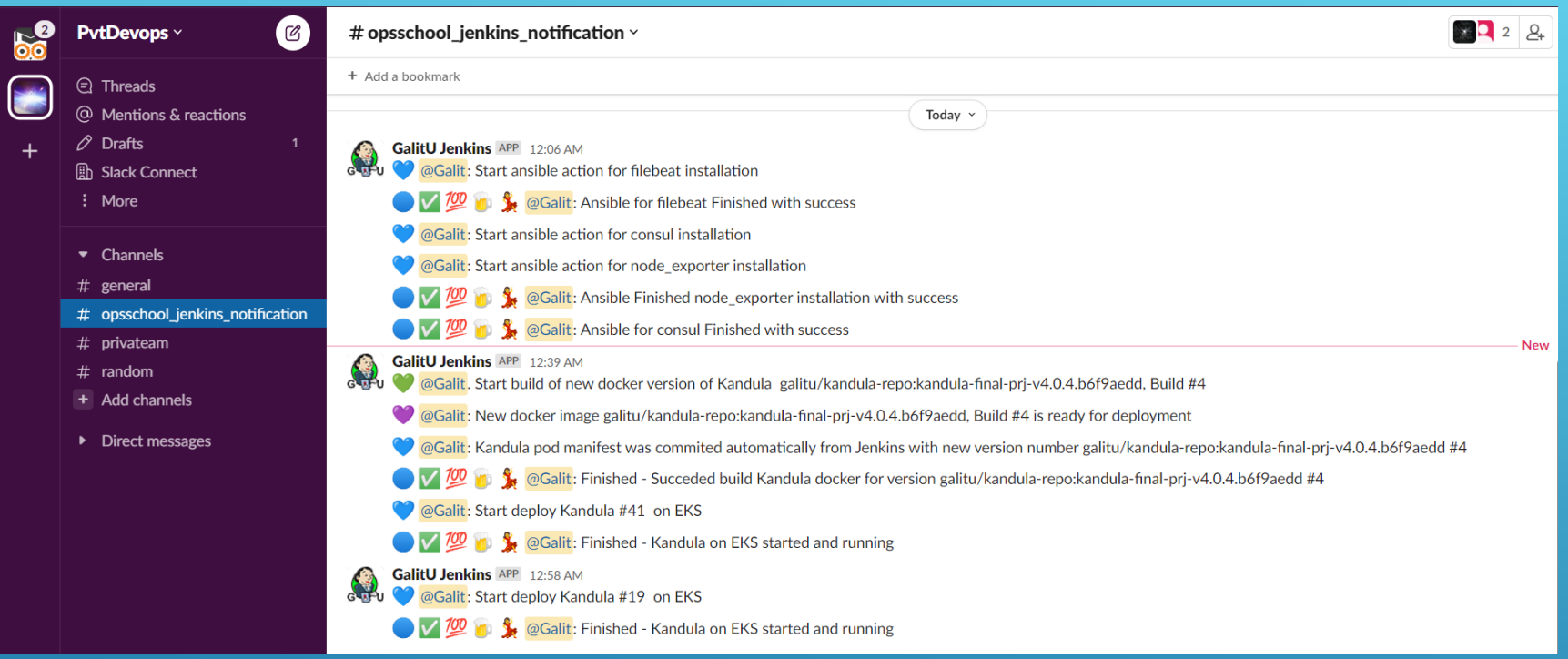
Options:
  -d, --debug          Add more debug info.
  -u, --url TEXT       Web address of site. Example :
                       https://kandula.galitu.sites  [required]
  --user TEXT          User name with which Kandula application is started
                       [required]
  -p, --password TEXT  Password with which Kandula application is started
                       [required]
  -i, --instance TEXT  Instance ID, can be multiple. Example: -i
                       i-078981293233d0fed -i i-0ab0fe580b92e2f26  [required]
  --help               Show this message and exit.
```







Slack notification system









Links to relevant git repo

- https://github.com/GalitU/Kandula_project
- https://github.com/GalitU/Kandula_deployment
- https://github.com/GalitU/Kandula_code
- https://github.com/GalitU/Ansible_for_jenkins







WORK CONDITIONS AND PROBLEMS

- I had 0 experience of cloud technology, so all what I learned was from scratch and took a time.
- I work full week at my office, and I am one women show, so there was a homework time between 21:00 – 05:00
- My regular working environment is Windows, “romance” with Unix is new for me and took to me some time at start
- There was many times that something what worked fine before just stopped to work, like Jenkins webhook, installation of ansible of consul… and I found my self make 1 step forward and 4 backward



WHAT I DECIDED NOT IMPLEMENT

- RDS + scheduler – I wrote many times software which used a DB including build a scheme and more. I have a proved experience with next list of DBs, so I decided to use a time for something new which I don't know
  - MySql
  - Posgress
  - LocalDB
  - SqlLite
- ELK – I started to implement if near the deadline and just not have a time to finish
- VPN – is a bonus, so decided not to implement
- Grafana alerts – I understood the idea, but it is taken an important time before end of project and decided not implement
- Pytest – I prepare unitests for many languages including python on my daily base at work so this is not new for me, even more I responsible for explain and teach developers at my work to do it, so not implemented because I have an experience at this







DECISIONS AFTER A COURSE

- All “not implemented” issues will be finished after the course
- I learned a lot and want to thank each of you and all for the opportunity to “touch” the sky (cloud)
- After the course I will implement at my work 2/5 of learned stuff which will be allowed by company security :☺
- Again ☺, there are a lot of things to learn, and now I have a base to start it



THANK YOU

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https://www.linkedin.com/in/galitudovik/?originalSubdomain=il

This presentation is at http://galitu.site/FinalOpsSchoolPresentation.pdf



